

## Python on the IBMi- Chapter 4



Our hero's journey has brought him far from the quiet forests of "Hello, World" to the twisting caves of pyodbc and SQL. But now, a new challenge looms. A vast, sticky labyrinth stretches before him, whispering with echoes of unseen threats and hidden knowledge. The Web.

In this chapter, we venture into the realm of web servers on the IBMi. For many including myself this part of the journey has always felt like wandering into a spider's den: tricky, unfamiliar, and a little intimidating. But fear not! With the right tools and a bit of courage, even this tangled domain can be conquered.

We'll use Python to spin up a simple web server on the IBMi and begin laying the groundwork for future adventures like fetching data from IBMi.

While tools like [ILEastic](#) ([ILEastic Article](#)) offer powerful magic for building microservices, this approach is refreshingly simple perfect for small utilities and personal quests. Just be sure Python is installed, and you've got the right open source gear in your pack.

So gather your courage, ready your gear, and step carefully the Web awaits.

## Disclaimer

This article will use the built-in Python HTTP Server. While this works, it is NOT production ready. While it might work for simple utilities and very low demand. It's a very simple HTTP Server that handles basic requests. The next few articles are for demonstration purposes only.

### Prerequisites

- Please refer to chapter 1 for other prerequisites (<https://www.linkedin.com/pulse/python-ibmi-chapter-1-ricky-thompson-zg7te/>)
- Code can be found here: ([https://github.com/thomprrl/ibmi\\_python\\_rpg\\_adventures](https://github.com/thomprrl/ibmi_python_rpg_adventures))

## Objective

- Start a Python web server and display a Hello World web page

## Let's Code

So I created a new folder called (**/home/rthompson/python/chapter4**). In this folder I created a simple web page named helloworld.html with the following content.

```
<!DOCTYPE html>
<html>
<body>
  <h1>Hello from Python and your IBMi</h1>
</body>
</html>
```

## Start the Web Server

Now let's start the web server. Make sure the port you're using (e.g., 43001) is available on your IBMi.

From VS Code Open a PASE terminal by pressing (**Ctrl+Shift+J**) and selecting PASE and then type the following:

```
cd python/chapter4
python3 -m http.server 43001
```

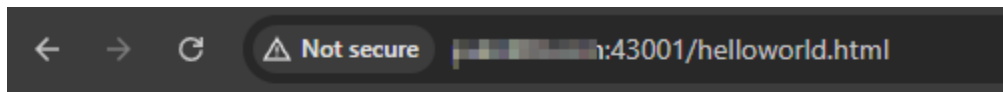
It will take a few seconds to start the server. You will see a message in the terminal that says something like Serving HTTP on 0.0.0.0 port 43001 (<http://0.0.0.0:43001>).

```
Terminal started, thanks for using Code for IBM i. Current directory is /home/RTHOMPSON.  
-bash-5.2$ cd python/webserver  
-bash-5.2$ python3 -m http.server 43001  
Serving HTTP on 0.0.0.0 port 43001 (http://0.0.0.0:43001/) ...  
|
```

Now open a web browser and go to your IBMi name or IP address plus the port number.

`http://myibmi:43001/helloworld.html`

You should see a page open up like the one below in the web browser.



## Hello from Python and your IBM i

To stop the web server, you can terminate the PASE terminal session by pressing Ctrl+C in the PASE terminal. Or kill the PASE Terminal.

### What's Next

With the web server now standing firm on the riverbank, our hero gazes across a wide, churning river. On the far side lies the ancient stronghold of RPG the place where wisdom and battle-hardened logic dwell. To reach it, he must lash together a raft of code and courage, and push off into unknown waters.

In Chapter 5, we'll build on our foundation by using the Python web server to call a backend RPG program. The goal? To retrieve data from deep within the IBMi.

Why lean on RPG? Because while our hero wields a new blade in Python, his shield of RPG knowledge and its proven business logic still serves him well.

So steady your raft, and brace yourself the current is strong, but the treasures ahead are worth the crossing.